PCB Board Design: A Complete Industry-Aligned Course

A 4-Month Practical Program for Early Professionals, Graduate Students & Career Transitioners

4 MONTHS | INDUSTRY-ALIGNED | PROJECT-BASED

Why This Course Matters

PCB Design Is at the Heart of Every Electronic Product

From smartphones to electric vehicles, satellites to medical devices — every electronic system depends on a well-designed printed circuit board. PCB design is one of the most in-demand, cross-disciplinary skills in modern electronics engineering, bridging hardware, firmware, signal integrity, and manufacturing.

🌍 Global Demand

The PCB market exceeds **$80B globally** with growing demand for skilled PCB designers in every major industry sector.

💼 Career Impact

PCB design proficiency is a **top hiring criterion** at aerospace, consumer electronics, automotive, and IoT companies worldwide.

🔧 Skill Gap

Most academic programs teach circuit theory but skip layout. This course fills that **critical industry gap** with hands-on practice.

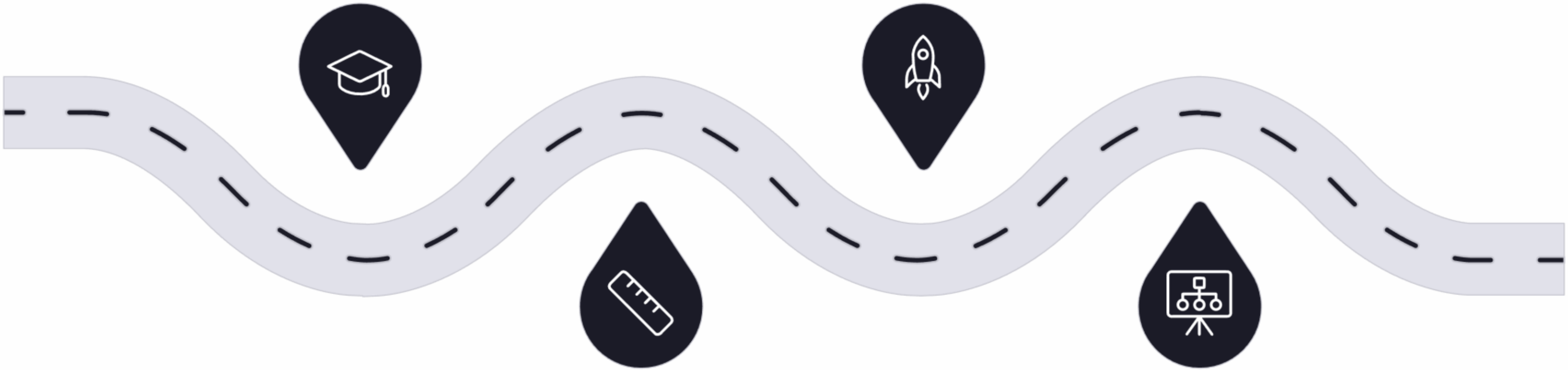
4-Month Structure at a Glance

Foundation

Weeks 1–4: basics and EDA tools

Advanced

Weeks 9–12: high-speed and advanced techniques



Intermediate

Weeks 5–8: layout and signal integrity

Capstone

Weeks 13–16: full board project and review

Each week includes a **weekend recorded session** for concept delivery and a **live 1:1 or small-group mentoring session** for doubt clearing, design feedback, and case discussions. The final 3–4 weeks are dedicated to a full capstone board design project reviewed by industry mentors.

Weekly Format

How Each Week Is Structured

📹 Weekend Recorded Session

- Concept-focused video content (2–3 hours)
- Tool walkthroughs and live design demos
- Annotated design files shared post-session
- Self-paced quizzes and knowledge checks
- Reading resources and industry documentation

🎯 Live Mentoring Session

- 1:1 or small-group doubt clearing (60–90 min)
- Design reviews and peer critique sessions
- Real-world case study discussions
- Assignment feedback and iteration guidance
- Industry Q&A and career coaching moments

Every learner receives a personal mentor for design feedback. Small group sizes (max 6 per cohort) ensure individual attention throughout the program.

PHASE 1

Foundation Phase — Weeks 1 to 4

Build the conceptual and tooling foundation required for every subsequent design decision. Weeks 1–4 cover electronics fundamentals, EDA tool setup, schematic capture, and design workflow basics.

1

Week 1: Electronics & PCB Fundamentals

- Role of PCBs in electronic systems
- PCB types: rigid, flex, rigid-flex, HDI
- Key components: resistors, capacitors, ICs, connectors
- PCB layer stackup basics and material selection
- Industry standards: IPC-2221, IPC-7351
- Overview of the end-to-end PCB design workflow

2

Week 2: EDA Tool Setup & Interface

- Introduction to KiCad, Altium Designer, and Eagle
- Tool installation, licensing, and workspace setup
- Understanding the schematic editor interface
- PCB editor navigation and layer management
- Library management: component and footprint libraries
- File formats: Gerber, ODB++, BOM, netlist

3

Week 3: Schematic Capture Essentials

- Schematic design rules and best practices
- Creating and editing components and symbols
- Net labeling, power flags, and bus structures
- Hierarchical schematics for large designs
- Running electrical rules checks (ERC)
- Generating netlists and bill of materials (BOM)

4

Week 4: PCB Design Workflow & DFM Basics

- Netlist import and board setup
- Defining board outline and mechanical constraints
- Placement strategy: grouping by function
- Introduction to Design for Manufacturability (DFM)
- Understanding manufacturer design rules (DRC)
- Mini Project: Schematic and layout of a basic LED circuit

PHASE 2

Intermediate Phase — Weeks 5 to 8

Elevate your designs from basic to production-ready. Weeks 5–8 cover component placement strategy, routing techniques, power distribution, and signal integrity fundamentals.

1

Week 5: Component Placement Strategy

- Placement rules: decoupling caps, thermal relief, orientation
- Functional block grouping and mechanical clearances
- Component reference designators and silkscreen management
- High-frequency vs. low-frequency placement zones
- Thermal management: hot spots, heat sinks, vias
- Design review checklist for placement phase

2

Week 6: Routing Techniques & Best Practices

- Manual vs. autorouting: when to use each
- Trace width calculation based on current load
- Differential pair routing and length matching
- Via types: through-hole, blind, buried, micro-via
- Ground planes and copper pours
- Avoiding routing pitfalls: acute angles, stubs, loops

3

Week 7: Power Distribution Network (PDN) Design

- PDN fundamentals: impedance, decoupling strategy
- Power plane design and copper pour management
- Voltage regulators: LDO vs. switching regulator placement
- Bypassing and bulk capacitor placement rules
- PDN simulation using free tools (PDN Analyzer)
- Common PDN failure modes and how to prevent them

4

Week 8: Signal Integrity Fundamentals

- Transmission line theory: impedance, reflection, crosstalk
- Controlled impedance traces and stackup planning
- Return path management and split plane avoidance
- EMI sources and layout-based mitigation techniques
- Signal integrity simulation tools: HyperLynx, SIwave
- Mini Project: 4-layer mixed-signal board with PDN design

PHASE 3

Advanced & Application Phase — Weeks 9 to 12

Apply professional-grade techniques used in aerospace, consumer electronics, and automotive PCB design. Weeks 9–12 cover high-speed design, HDI, rigid-flex, and the full manufacturing handoff workflow.

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Week 9: High-Speed PCB Design

- High-speed design rules: DDR4, PCIe, USB 3.x, Ethernet
- Length matching, skew control, and serpentine routing
- Layer stackup optimization for high-speed signals
- Connector and via transitions at high frequencies
- Pre-layout and post-layout SI simulation workflows
- Reading and interpreting eye diagrams

2

Week 10: HDI & Advanced Packaging

- HDI board classification: 1+N+1, 2+N+2 stackups
- Microvias, stacked and staggered via design
- BGA fanout strategies and escape routing
- Fine-pitch component design rules
- Cost implications of HDI: when is it justified?
- Industry use case: smartphone main board analysis

3

Week 11: Rigid-Flex & Specialized PCB Types

- Rigid-flex construction: zones, bend radius, material stack
- Flex circuit design rules and conductor orientation
- Coverlay vs. solder mask in flex regions
- RF PCB design: microstrip, coplanar waveguide, shielding
- High-power PCB design and thermal management
- Medical and aerospace design compliance considerations

4

Week 12: Manufacturing Handoff & DFM/DFA

- Generating complete Gerber and drill files
- Assembly drawings, fabrication notes, and BOM preparation
- DFM audit: panelization, fiducials, solder mask requirements
- DFA checklist: component orientation, polarity marking
- Working with CM (contract manufacturers) and DFM reviews
- Mini Project: Full manufacturing output package for 6-layer board

PHASE 4

Capstone Project — Weeks 13 to 16

The capstone is a fully scoped, end-to-end PCB design project simulating an industry assignment. Learners choose from defined project tracks and deliver a complete, manufacture-ready board package reviewed by a senior industry mentor.

Project Tracks

- 🏠 IoT Smart Home Controller (4-layer)
- ⚡ Motor Driver Power Board (high-current)
- 📡 Wireless Communication Module (RF + BLE)
- 🏥 Portable Medical Sensor Board

Deliverables Required

- **Week 13:** Schematic review and design architecture sign-off
- **Week 14:** Layout review — placement, routing, and stackup
- **Week 15:** DRC/ERC clean, SI checks, and peer review
- **Week 16:** Full manufacturing package + recorded design walkthrough

Capstone boards are evaluated on IPC compliance, signal integrity, DFM quality, and design documentation standards.

Industry Example 1

Redesigning an IoT Gateway for Cost and Signal Compliance

🔍 Business Problem

A mid-sized industrial IoT company was experiencing **15% field return rates** on its factory-floor gateways due to EMI failures during FCC/CE certification. The 2-layer design had inadequate ground planes and poor RF isolation, causing radiated emissions to exceed limits.

⚙️ How PCB Design Was Applied

The team migrated to a **4-layer stackup** with solid ground and power planes, implemented controlled-impedance traces for the Wi-Fi/BLE antenna path, added copper pour stitching vias, and isolated the switching regulator from the RF section using a physical keepout zone.

🛠️ Tools Used

Altium Designer for layout, HyperLynx for pre-layout EMI simulation, Saturn PCB Toolkit for trace impedance, and Ansys SIwave for final PDN validation.

📈 Business Impact & Early Professional Role

Field return rate dropped to **under 1%**. Time-to-certification reduced by 6 weeks. Early professionals contributed to stackup definition, DRC reviews, and Gerber file preparation — core skills taught in Weeks 6–12 of this course.

High-Speed DDR4 Memory Interface for an Edge AI Board

🔍 Business Problem

A startup developing an edge AI inference board needed to route a **64-bit DDR4 memory interface** at 3200 MT/s between a custom SoC and four DRAM chips on a compact 6-layer board. Initial prototypes suffered from bit error rates that made the system unreliable at full speed.

⚙️ How PCB Design Was Applied

Engineers applied **fly-by topology** for command/address signals, length-matched all data byte lanes within ±5 mil, implemented a reference plane directly beneath all DDR4 signal layers, and used proper via back-drilling to eliminate via stubs at high frequencies.

🛠️ Tools Used

Cadence Allegro for routing with constraint-driven design rules, Sigrity PowerSI for PDN analysis, and oscilloscope eye-diagram measurements at board bring-up for validation.

📈 Business Impact & Early Professional Role

Bit error rate dropped to acceptable levels at full DDR4-3200 speed. Board passed first-pass bring-up. Junior engineers executed the **length-matching, DRC setup, and manufacturing output** tasks — directly mapped to Weeks 9 and 12 of this course.

Medical-Grade Wearable Sensor PCB — Compliance & Miniaturization

🔍 Business Problem

A digital health company was designing a **continuous glucose monitoring wearable**. The board had to pass IEC 60601-1 medical electrical safety standards, fit within a 15×20mm form factor, survive skin-contact flex bending cycles, and achieve 7-day battery life — all simultaneously.

⚙️ How PCB Design Was Applied

The design used a **rigid-flex PCB construction** with two rigid zones and a single flex interconnect. HDI microvias enabled BGA escape routing on the ASIC. Ultra-low-leakage components were placed per IEC creepage/clearance tables, and the battery management IC was isolated from the analog front-end.

🛠️ Tools Used

Altium Designer with rigid-flex layer definitions, Zuken CR-8000 for 3D mechanical constraint verification, and IPC-2223 as the reference standard for flex design rules.

📈 Business Impact & Early Professional Role

Device cleared IEC 60601-1 on the first submission — saving an estimated **$120K in re-testing costs**. Early professionals managed component research, footprint creation, and DFM checklist execution — all covered in Weeks 10–12 of this program.

Tools & Technology Stack

The PCB Designer's Professional Toolkit

Proficiency across multiple tools is expected in industry roles. This course focuses on **KiCad** as the primary free tool for all exercises, with structured exposure to Altium Designer and Eagle. Simulation and analysis tools are introduced in the Advanced Phase.

EDA / Layout Tools

- **KiCad 7** — Primary course tool (free, open-source)
- **Altium Designer** — Industry standard (trial license)
- **Cadence Allegro** — Enterprise-grade (intro exposure)
- **Eagle** — Widely used in maker/startup environments

Simulation & Analysis

- **Saturn PCB Toolkit** — Trace width, impedance calc
- **HyperLynx** — SI/PI pre-layout simulation
- **PDN Analyzer (KiCad)** — Power distribution analysis
- **LTspice** — Circuit-level simulation and verification

Manufacturing & Collaboration

- **JLCPCB / PCBWay** — Online fab for prototyping
- **GerbView / FAB 3000** — Gerber file verification
- **GitHub** — Design version control
- **Notion / Confluence** — Design documentation

Career & Industry Alignment

Where This Course Takes Your Career

PCB design skills map directly to high-demand roles across electronics manufacturing, automotive, aerospace, consumer tech, and medical device industries. Below are the primary roles this course prepares you for.

🖥️ PCB Layout Engineer

Executes board layouts from schematic to manufacturing outputs. Core skills: multi-layer routing, DRC/ERC, Gerber generation. **Avg. Salary: $75K–$110K**

⚡ Hardware Design Engineer

Owns schematic design and PCB architecture decisions. Requires SI, PDN, and cross-functional team collaboration. **Avg. Salary: $90K–$130K**

📡 Signal Integrity Engineer

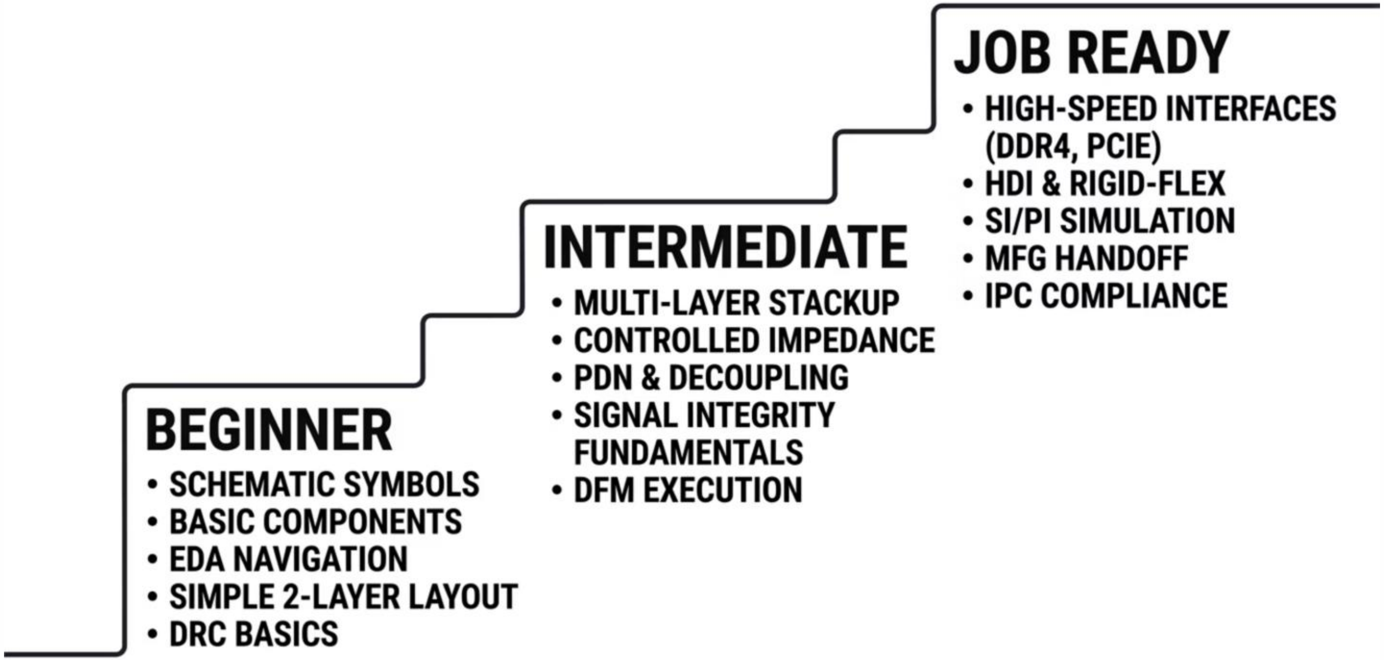
Specializes in high-speed interface validation, simulation, and post-silicon debug. Advanced path requiring strong SI fundamentals. **Avg. Salary: $110K–$150K**

🏭 DFM / Manufacturing Engineer

Bridges design and production, auditing boards for manufacturability and yield. Strong knowledge of IPC standards required. **Avg. Salary: $80K–$115K**

Skills Progression Map

From Beginner to Job-Ready: Your Skills Journey



Skills are built progressively — each phase unlocks the next. By the end of Week 12, learners will have demonstrated all **Job Ready** competencies through graded projects and a capstone design reviewed by an industry mentor.

What You Leave This Course With

A Professional Portfolio

Three graded mini-projects plus a full capstone board — a manufacture-ready portfolio that demonstrates real design competency to hiring managers and technical interviewers.

Multi-Tool Proficiency

Hands-on experience in KiCad, Altium Designer, SI simulation tools, and PDN analyzers — the exact stack expected in entry-level and mid-level PCB roles across industries.

Industry-Grade Practices

Deep familiarity with IPC standards, DFM/DFA principles, controlled-impedance design, and full manufacturing handoff — knowledge that directly reduces time-to-competency in a new job role.

A Clear Career Roadmap

A structured learning path that aligns with real job descriptions, plus access to mentor networks, peer communities, and ongoing design challenges to continue growing post-course.

Graduates receive a **course completion certificate** and access to the alumni design community — a network of PCB designers across industry roles for ongoing mentorship and referrals.

Post-Course Roadmap

Your Path After Completing the Program

Polish Portfolio

Refine capstone; document projects on GitHub

Certifications

Pursue IPC-A-600, IPC-7711, or Altium ACD

Apply & Interview

Target PCB Layout & Hardware roles; use referrals

First 90 Days

Join board reviews, shadow SI, learn DFM standards

Specialization

Choose High-Speed SI, RF, Medical, or Automotive ADAS

Recommended Certifications

- IPC-A-600 — Acceptability of Printed Boards
- IPC-7711/7721 — Rework and Repair
- Altium Certified Designer (ACD)
- Cadence Allegro Certification

Specialization Tracks to Explore

- High-Speed Digital & SerDes Design
- RF & Microwave PCB Engineering
- Automotive ADAS & ADAS-grade PCB
- Medical Device Board Design (IEC 60601)

Ready to Build the Boards That Power the World?

This course is your structured, mentor-supported, industry-aligned pathway into one of electronics engineering's most valuable and in-demand disciplines. Whether you're starting fresh, transitioning careers, or deepening existing skills — every lesson, project, and mentoring session is designed to move you closer to job-ready confidence.

📅 4 Months Structured, phased learning with clear weekly outcomes	🎯 3 Mini Projects Graded, mentor-reviewed real design exercises
🛠️ 1 Capstone Board Manufacture-ready design for your portfolio	💼 Job-Ready Skills Mapped to industry roles and real hiring criteria

Enroll today and start designing the future — one layer at a time.